

# Data Engineering 101

## Dimensional Modelling

# Steps Involved



## ACTIVITY

# IDENTIFY STAKEHOLDERS

Retail store managers, financial analysts, marketing team

## POSSIBLE PROBLEMS

# CONFLICTING PRIORITIES

Hold a stakeholder meeting to align goals



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## ACTIVITY

# GATHER REQUIREMENTS

Analyze sales data, track inventory levels, monitor customer behavior, evaluate marketing campaigns

## POSSIBLE PROBLEMS

# INCOMPLETE REQUIREMENTS

Conduct follow-up interviews with stakeholders



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## ACTIVITY

### DEFINE SCOPE

Specify the scope of the project including data sources and reporting needs

## POSSIBLE PROBLEMS

### SCOPE CREEP

Implement a change control process ✨



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## ACTIVITY

# LIST BUSINESS PROCESSES

Sales transactions, inventory management, customer management, marketing campaigns

## POSSIBLE PROBLEMS

# OVERLAPPING PROCESSES

Clarify and document each process



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## ACTIVITY

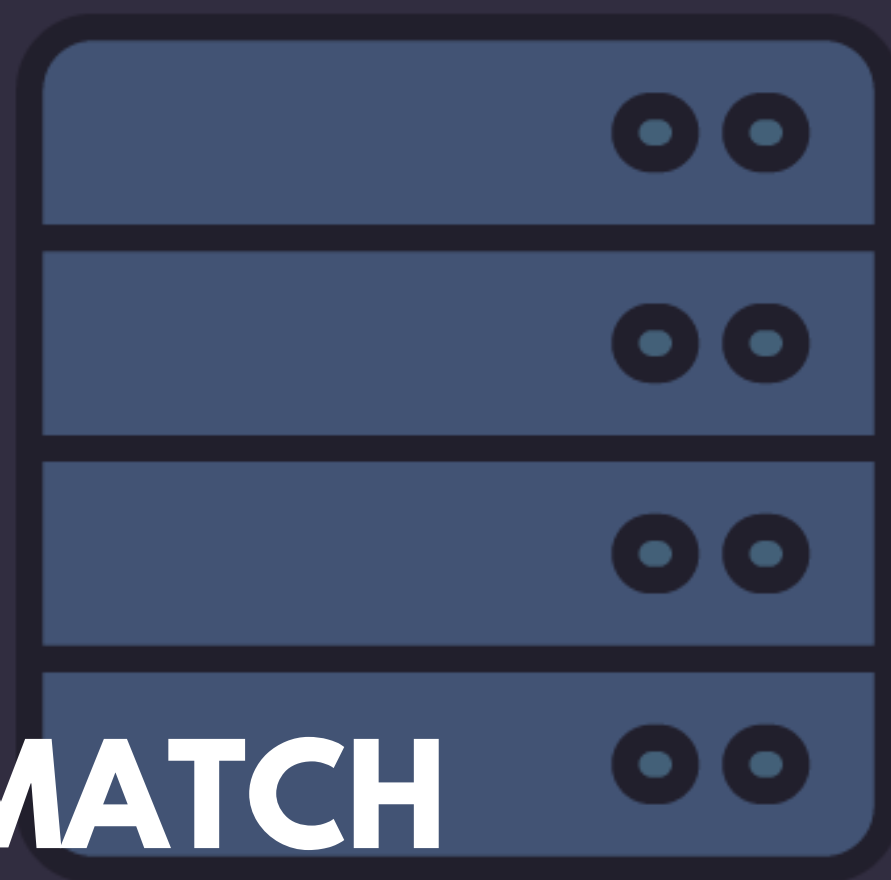
### DEFINE GRANULARITY

Transaction-level data, daily inventory levels, customer interactions

## POSSIBLE PROBLEMS

### GRANULARITY MISMATCH

Standardize granularity for consistent analysis



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## ACTIVITY

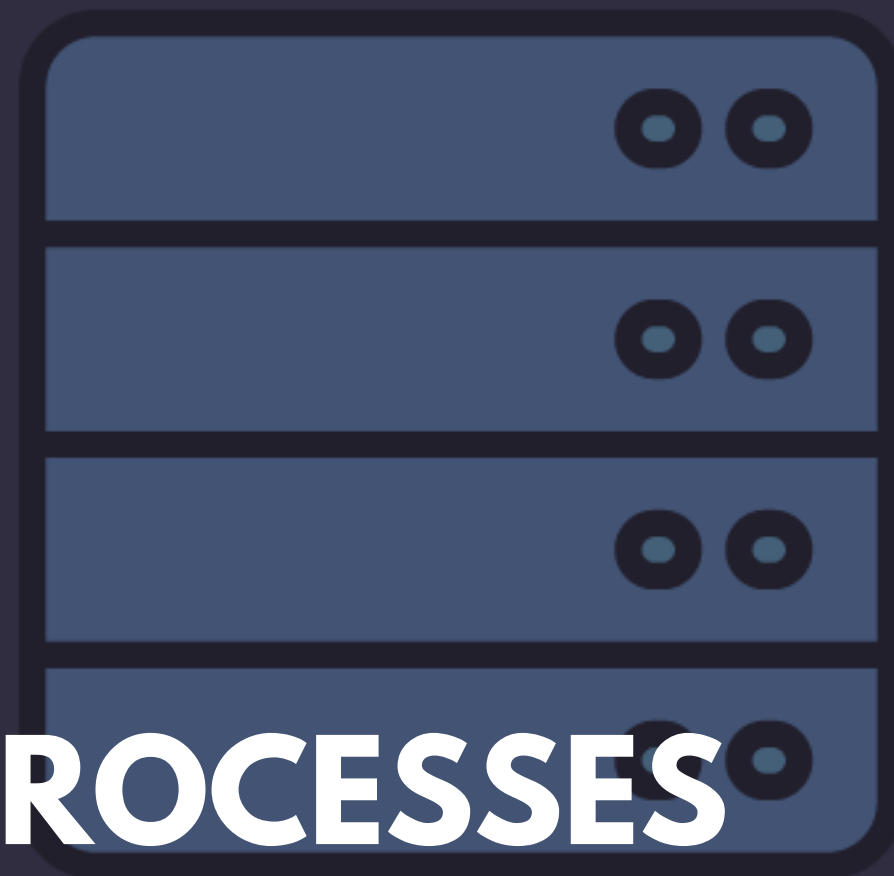
# DOCUMENT PROCESS FLOWS

Create flowcharts for each business process

## POSSIBLE PROBLEMS

# MISUNDERSTOOD PROCESSES

Review flowcharts with stakeholders for validation



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## ACTIVITY

# IDENTIFY DIMENSIONS

Product, Store, Customer, Time,  
Promotion

## POSSIBLE PROBLEMS

# MISSING DIMENSIONS

Revisit business requirements ✨



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## ACTIVITY

# DEFINE ATTRIBUTES FOR EACH DIMENSION

Product: ProductID, Name, Category, Price;

Store: StoreID, Location, Manager;

Customer: CustomerID, Name, Address, Age;

Time: Date, Day, Month, Year;

Promotion: PromotionID, Type, StartDate, EndDate



## POSSIBLE PROBLEMS

# ATTRIBUTE REDUNDANCY

Normalize attributes to reduce redundancy

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## ACTIVITY

# CREATE DIMENSION TABLES

Develop SQL scripts for creating dimension tables

## POSSIBLE PROBLEMS

# DATA TYPE ISSUES

Validate data types against source systems



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## ACTIVITY

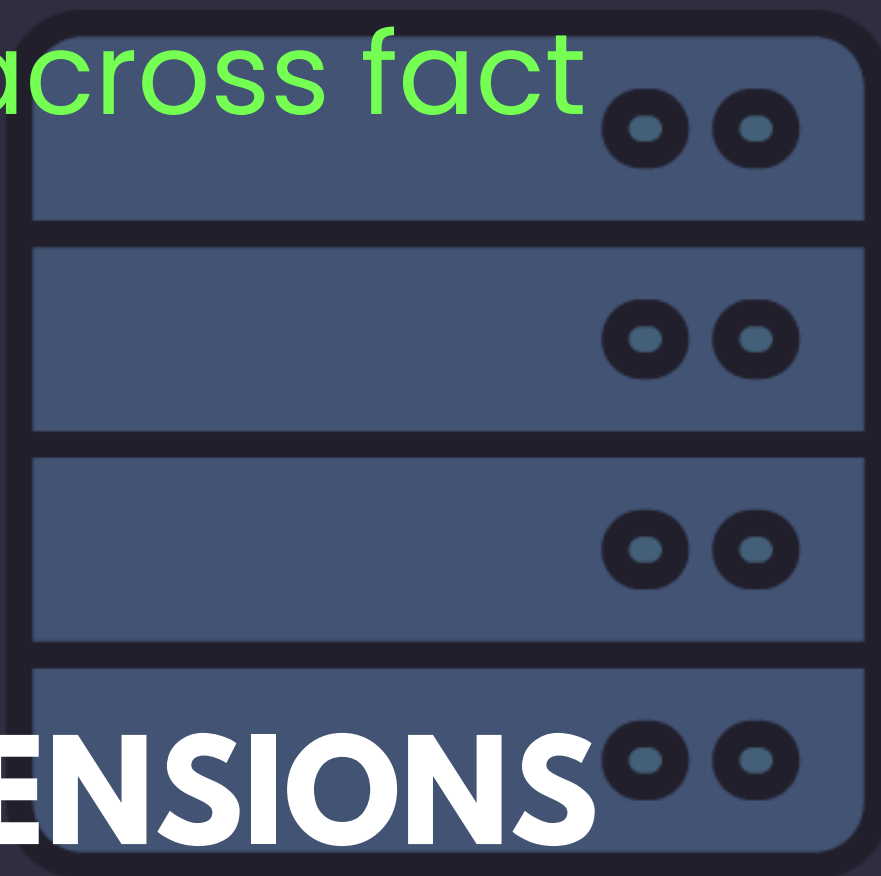
# IMPLEMENT CONFORMED DIMENSIONS

Ensure dimensions like Time and Product are consistent across fact tables

## POSSIBLE PROBLEMS

# INCONSISTENT DIMENSIONS

Standardize dimension tables and ensure consistency



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## ACTIVITY

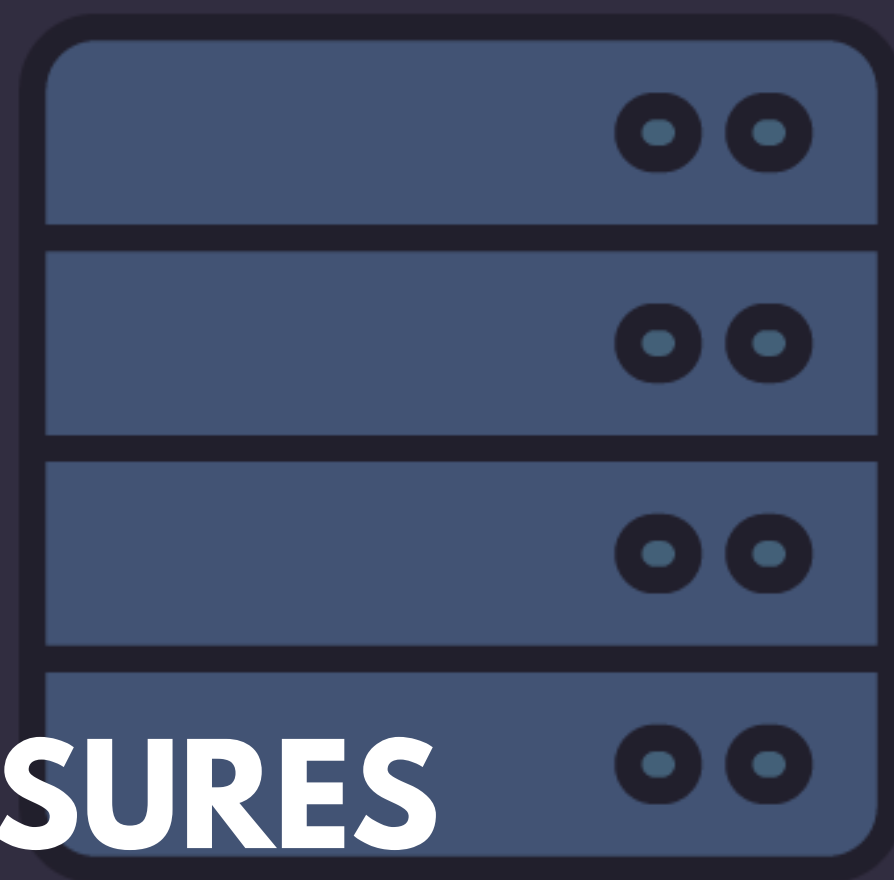
# IDENTIFY FACT TABLES

Sales, Inventory, Customer  
Interaction

## POSSIBLE PROBLEMS

# OVERLAPPING MEASURES

Clearly define the purpose of each  
fact table



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## ACTIVITY

### DEFINE MEASURES

Sales: SalesAmount, QuantitySold;

Inventory: StockLevel, ReorderLevel;

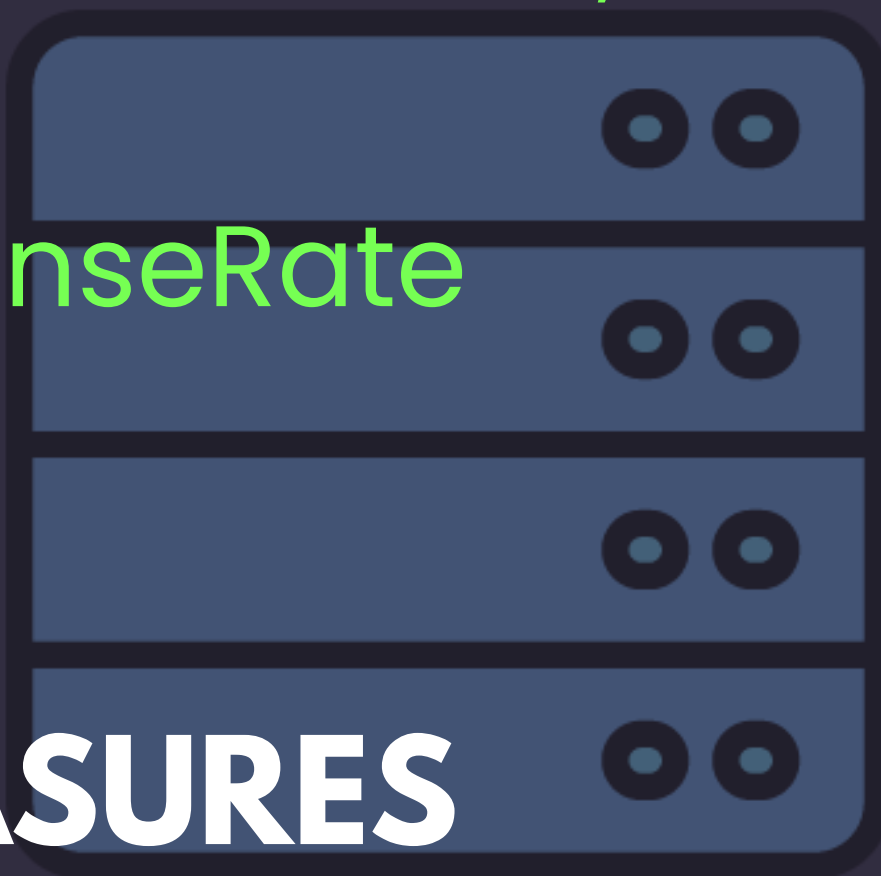
Customer Interaction:

InteractionCount, ResponseRate

## POSSIBLE PROBLEMS

### INCONSISTENT MEASURES

Standardize measurement units



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## ACTIVITY

# CREATE FACT TABLES

Develop SQL scripts for creating fact tables

## POSSIBLE PROBLEMS

# PERFORMANCE ISSUES

Optimize table structures and indexing



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## ACTIVITY

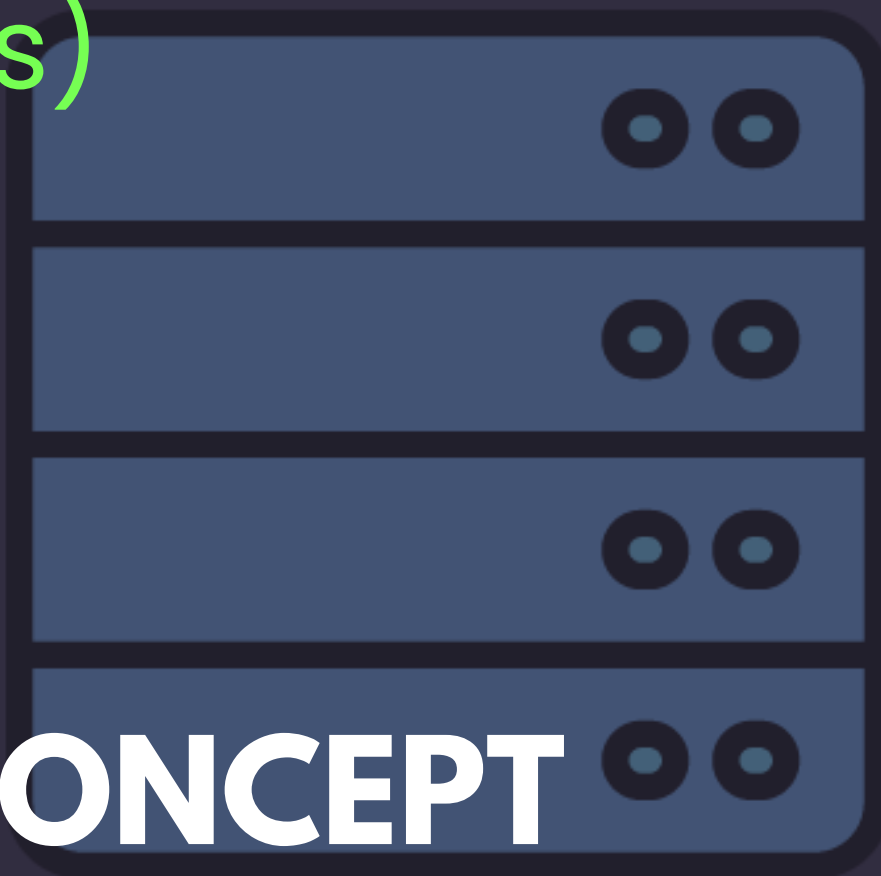
# DESIGN FACTLESS FACT TABLES

Capture events without measures  
(e.g., promotional events)

## POSSIBLE PROBLEMS

# MISUNDERSTOOD CONCEPT

Educate stakeholders on the importance of factless fact tables



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## ACTIVITY

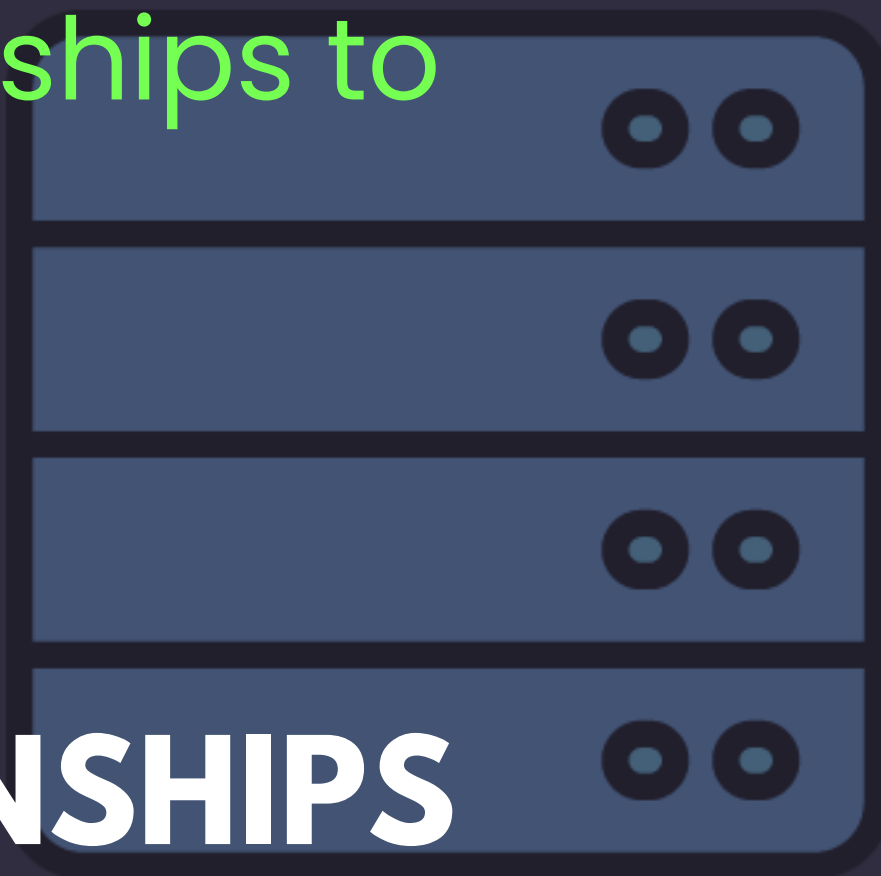
# CREATE STAR SCHEMA DIAGRAMS

Design diagrams showing fact  
tables and their relationships to  
dimension tables

## POSSIBLE PROBLEMS

# COMPLEX RELATIONSHIPS

Simplify by breaking into smaller  
diagrams



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## ACTIVITY

# ENSURE FOREIGN KEY RELATIONSHIPS

Link fact tables to dimension tables  
using foreign keys

## POSSIBLE PROBLEMS

# MISSING FOREIGN KEYS

Review and add necessary foreign  
keys



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## ACTIVITY

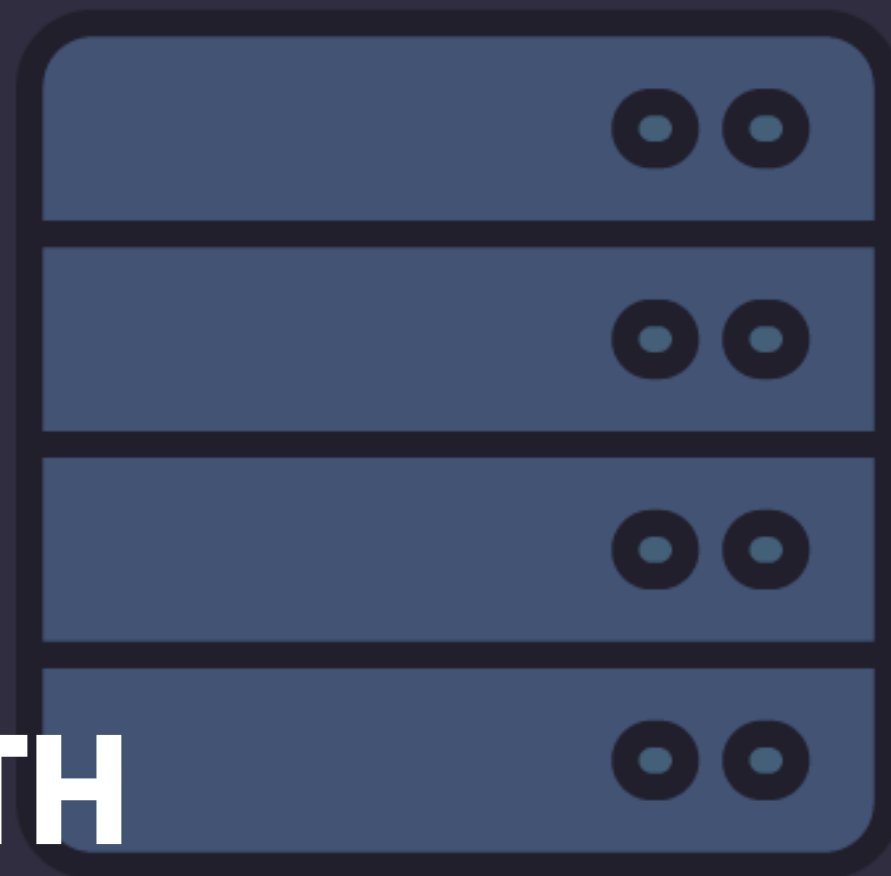
# VALIDATE SCHEMA DESIGN

Validate the star schema design  
with stakeholders

## POSSIBLE PROBLEMS

# MISALIGNMENT WITH REQUIREMENTS

Iterate design based on feedback



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## ACTIVITY

# IMPLEMENT DATE DIMENSION

Create a date dimension to handle various date attributes

## POSSIBLE PROBLEMS

# DATE HANDLING ISSUES

Standardize on a single date dimension table ✨



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## ACTIVITY

# IMPLEMENT SURROGATE KEYS

Assign surrogate keys to maintain unique identities for dimension records

## POSSIBLE PROBLEMS

# NATURAL KEY CHANGES

Use surrogate keys to avoid issues with changing natural keys



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## ACTIVITY

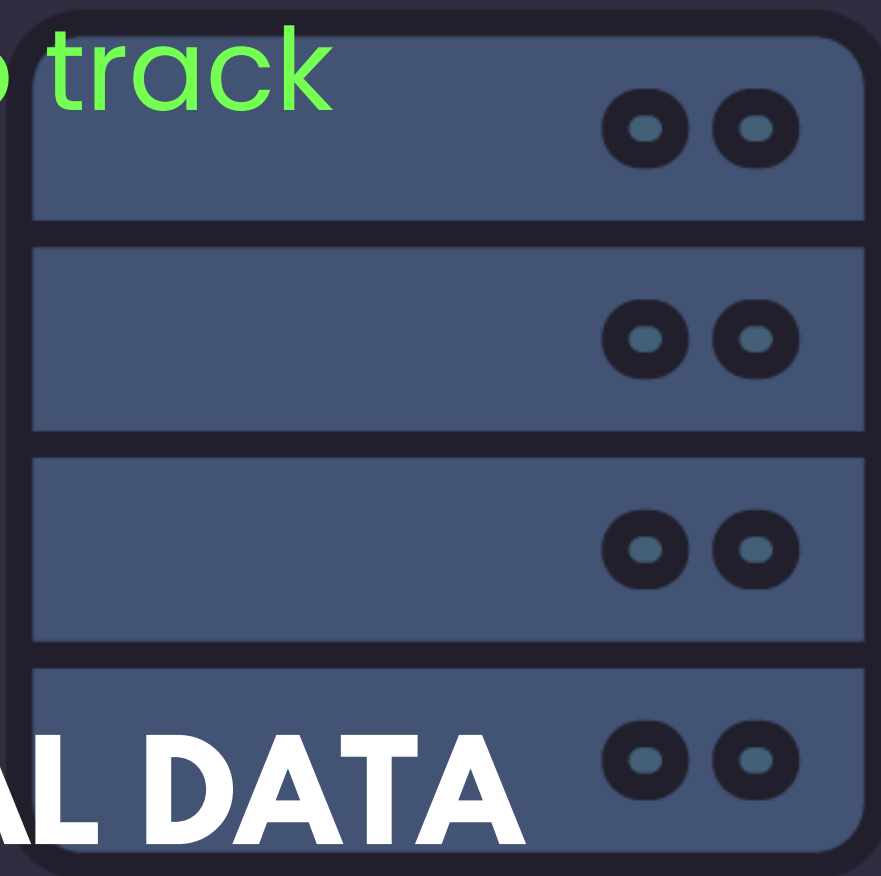
# IMPLEMENT SLOWLY CHANGING DIMENSIONS (SCD)

Implement SCD types to track  
historical changes

## POSSIBLE PROBLEMS

# LOSS OF HISTORICAL DATA

Use Type 1, Type 2, and Type 3 as  
appropriate



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## ACTIVITY

# HANDLE LATE ARRIVING DIMENSIONS

Process dimensions that arrive after fact data

## POSSIBLE PROBLEMS

### MISALIGNED DATA

Implement staging areas to temporarily hold late-arriving data ✨



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## ACTIVITY

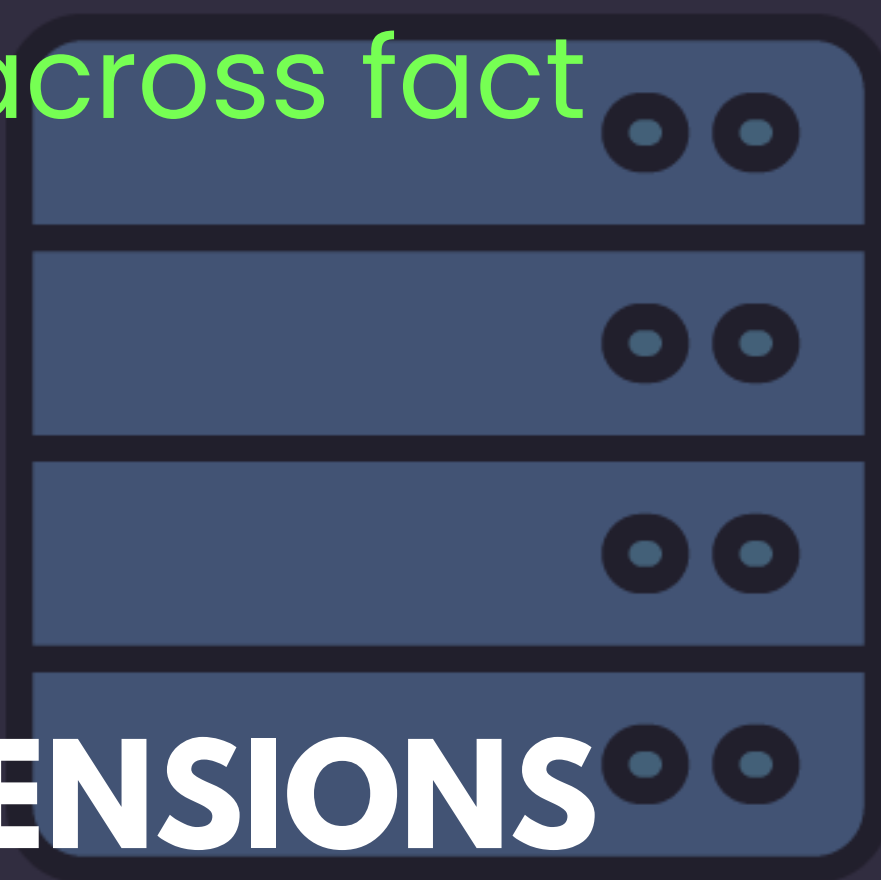
# IMPLEMENT CONFORMED DIMENSIONS

Ensure dimensions like Time and Product are consistent across fact tables

## POSSIBLE PROBLEMS

# INCONSISTENT DIMENSIONS

Standardize dimension tables and ensure consistency



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## ACTIVITY

# HANDLE MULTI-VALUED DIMENSIONS

Manage dimensions with multiple values (e.g., multiple product categories)

## POSSIBLE PROBLEMS

# COMPLEX QUERIES

Use bridge tables or flatten the structure



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## ACTIVITY

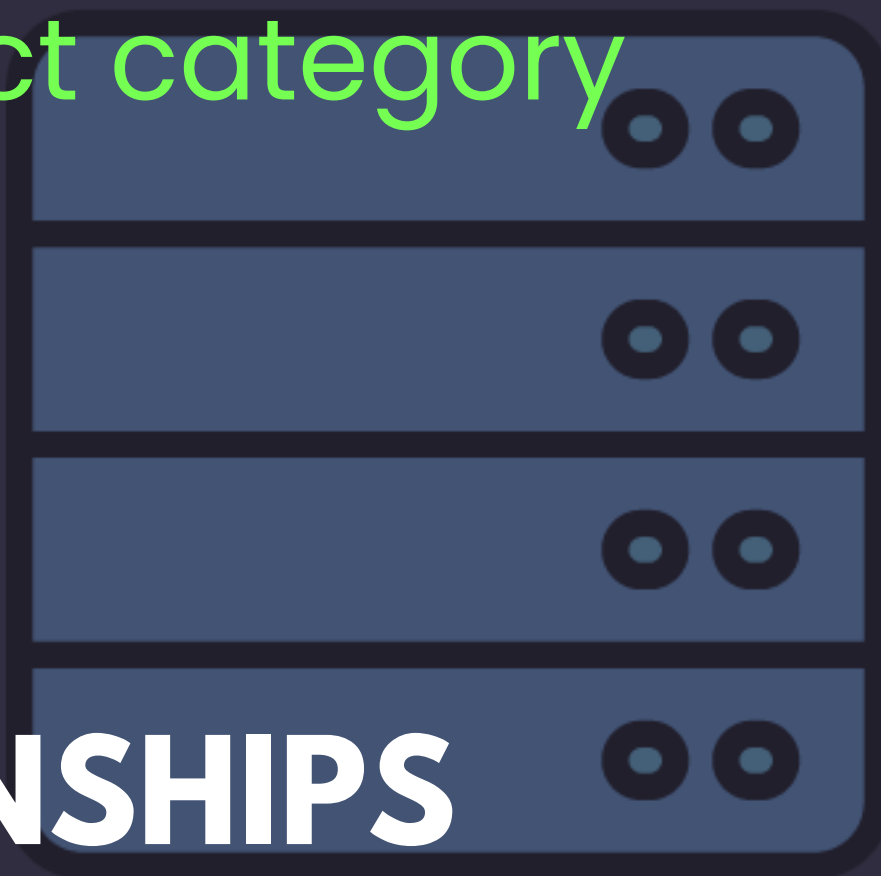
# DESIGN HIERARCHIES AND SNOWFLAKES

Design hierarchies within dimensions (e.g., product category hierarchy)

## POSSIBLE PROBLEMS

# COMPLEX RELATIONSHIPS

Use snowflake schema design where necessary



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## ACTIVITY

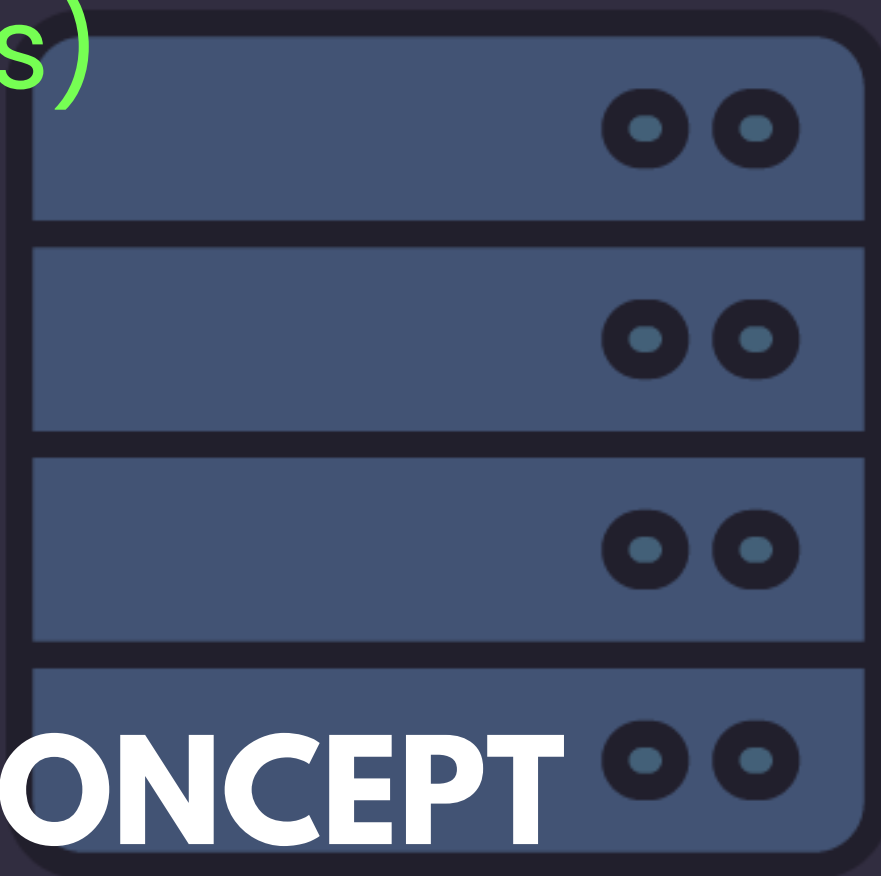
# IMPLEMENT FACTLESS FACT TABLES

Capture events without measures  
(e.g., promotional events)

## POSSIBLE PROBLEMS

# MISUNDERSTOOD CONCEPT

Educate stakeholders on the importance of factless fact tables



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## ACTIVITY

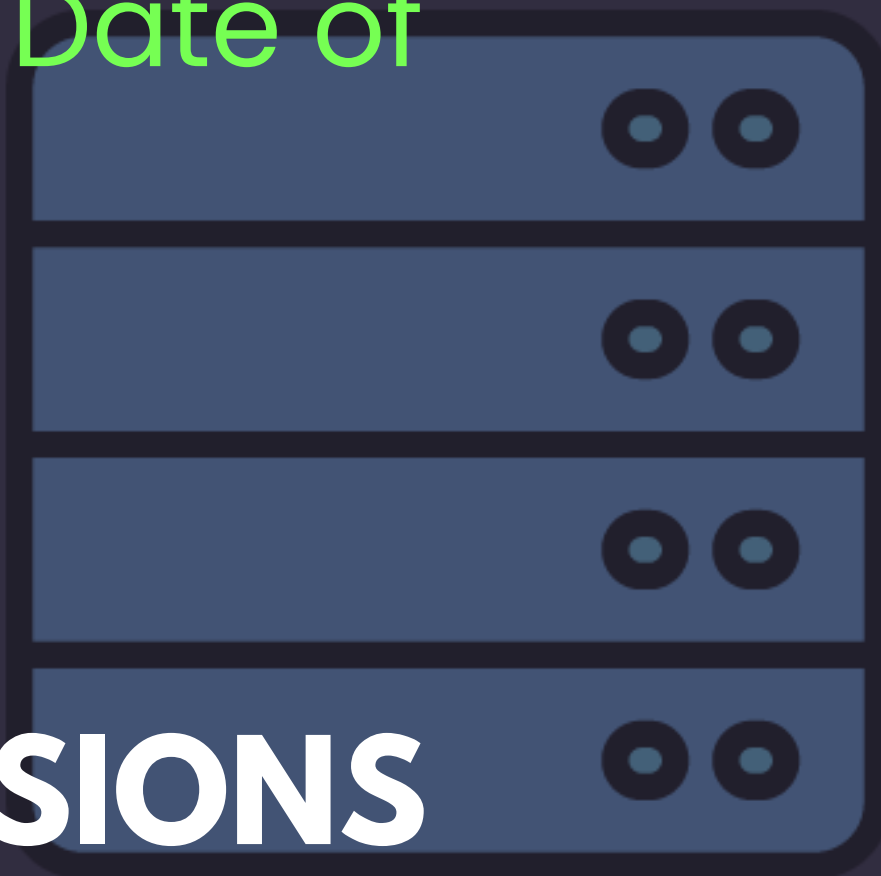
### DESIGN ROLE-PLAYING DIMENSIONS

Implement dimensions that can play multiple roles (e.g., Date of Sale, Date of Delivery)

## POSSIBLE PROBLEMS

### REDUNDANT DIMENSIONS

Create views to represent role-playing dimensions



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## ACTIVITY

# IMPLEMENT DEGENERATE DIMENSIONS

Capture transactional identifiers without corresponding dimension tables

## POSSIBLE PROBLEMS COMPLEX QUERIES

Use degenerate dimensions for invoice numbers, transaction IDs



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## ACTIVITY

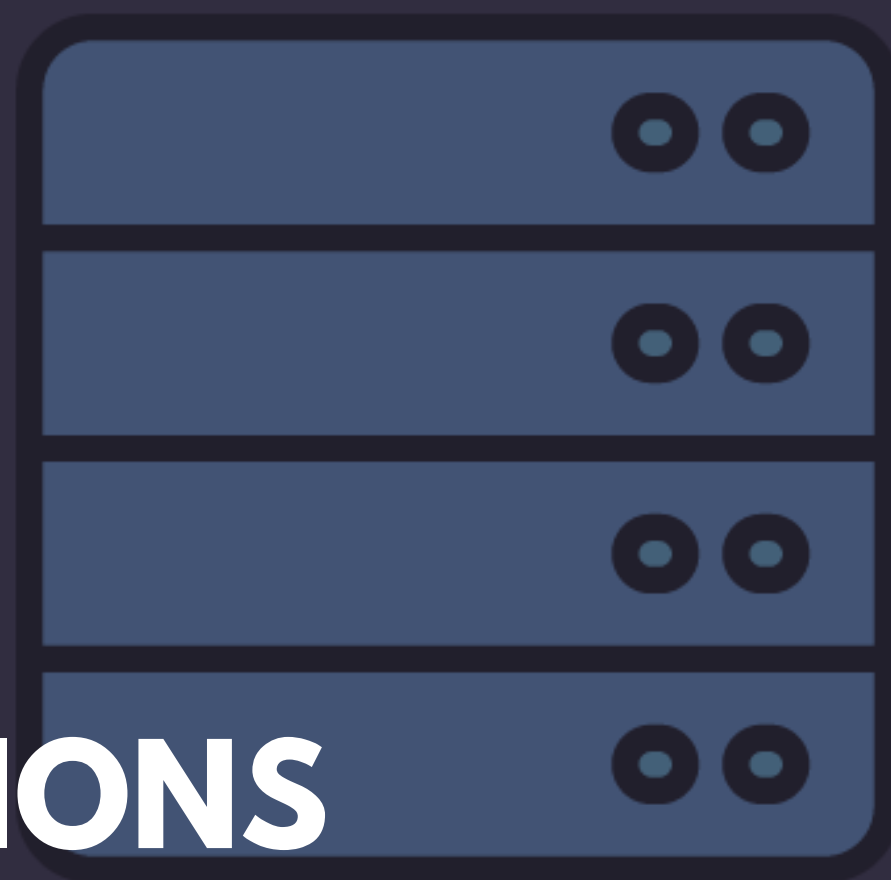
### DESIGN JUNK DIMENSIONS

Consolidate low-cardinality flags and indicators into a single dimension

## POSSIBLE PROBLEMS

### EXCESSIVE DIMENSIONS

Combine attributes into a single junk dimension



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## ACTIVITY

# HANDLE RAGGED HIERARCHIES

Implement hierarchies with uneven levels (e.g., geographic hierarchies)

## POSSIBLE PROBLEMS

# COMPLEX STRUCTURES

Use bridge tables to manage  
ragged hierarchies



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## ACTIVITY

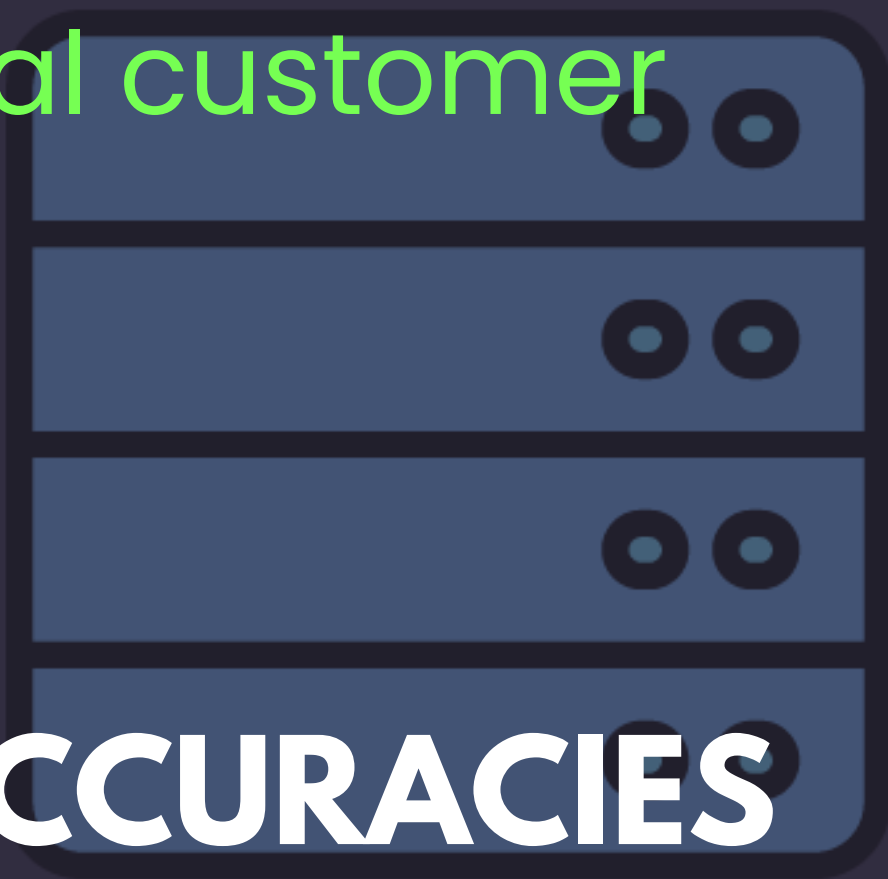
### IDENTIFY COMMON AGGREGATES

Monthly sales, quarterly sales by product category, annual customer growth

## POSSIBLE PROBLEMS

### AGGREGATION INACCURACIES

Implement validation checks for aggregate data



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## ACTIVITY

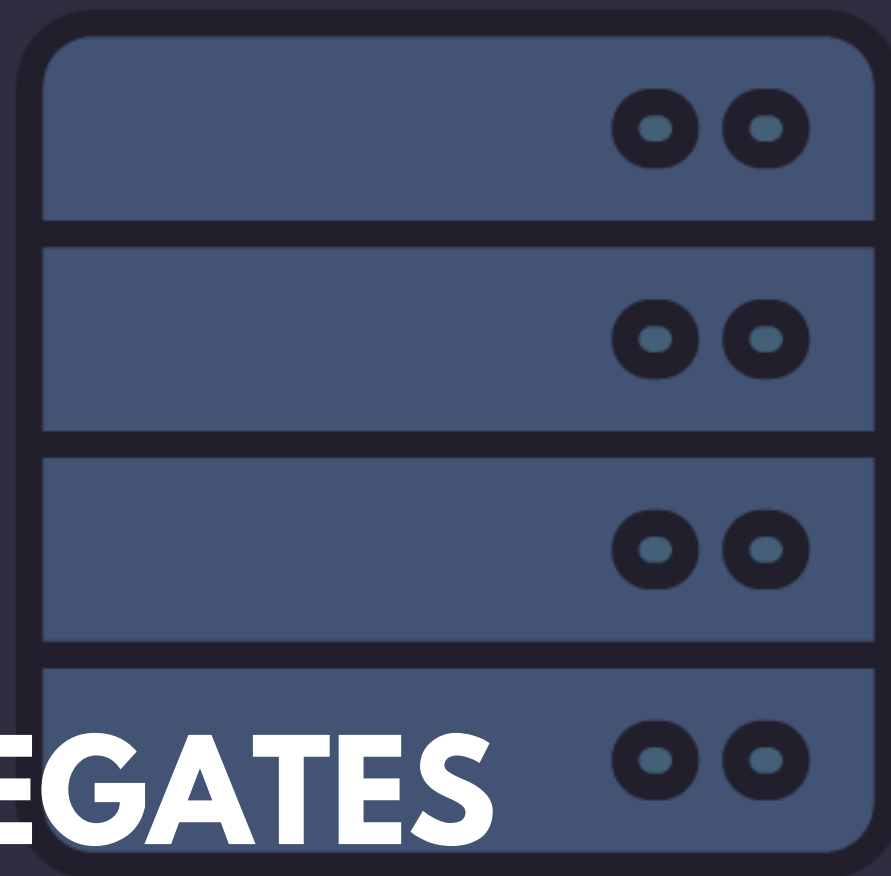
# DESIGN AGGREGATE TABLES

Create summary tables to store aggregated data

## POSSIBLE PROBLEMS

# REDUNDANT AGGREGATES

Ensure aggregates add business value



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## ACTIVITY

# IMPLEMENT AGGREGATE TABLES

Develop SQL scripts to create and populate aggregate tables

## POSSIBLE PROBLEMS

# PERFORMANCE ISSUES

Index aggregate tables for faster access



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## ACTIVITY

# DESIGN SNAPSHOT FACT TABLES

Capture data at regular intervals  
(e.g., end-of-month balances)

## POSSIBLE PROBLEMS

### DATA VOLATILITY

Use snapshot tables for periodic data



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## ACTIVITY

# DESIGN ACCUMULATING SNAPSHOT FACT TABLES

Track the lifecycle of an event (e.g.,  
order processing)

## POSSIBLE PROBLEMS

# COMPLEX LIFECYCLE EVENTS

Use accumulating snapshots for  
tracking progress



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## ACTIVITY

# IMPLEMENT LATE-ARRIVING FACTS

Process fact records that arrive late

## POSSIBLE PROBLEMS

### DATA ALIGNMENT ISSUES

Adjust fact processing to handle late-arriving records



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## ACTIVITY

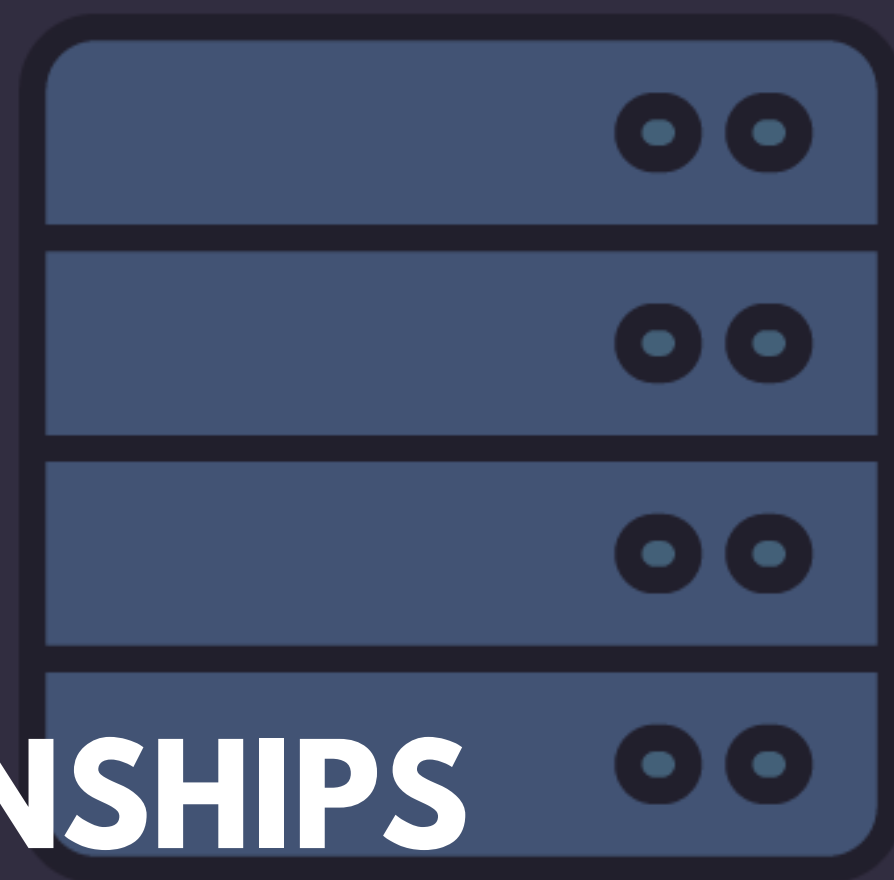
# IMPLEMENT BRIDGE TABLES

Manage many-to-many relationships (e.g., multiple roles for an employee)

## POSSIBLE PROBLEMS

# COMPLEX RELATIONSHIPS

Use bridge tables to resolve many-to-many relationships



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## ACTIVITY

# OPTIMIZE DIMENSION TABLES

Index dimension tables for faster query performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Create appropriate indexes



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## ACTIVITY

# OPTIMIZE FACT TABLES

Index fact tables and partition for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing and partitioning strategies ✨



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## ACTIVITY

# MONITOR AND MAINTAIN DIMENSIONS

Regularly update and maintain dimension tables

## POSSIBLE PROBLEMS

### DATA STALENESS

Implement update processes for dimensions



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## ACTIVITY

# MONITOR AND MAINTAIN FACT TABLES

Regularly update and maintain fact tables

## POSSIBLE PROBLEMS

# DATA STALENESS

Implement update processes for facts



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## ACTIVITY

# IMPLEMENT DATA LINEAGE TRACKING

Track the origin and transformation of data

## POSSIBLE PROBLEMS

# DATA TRACEABILITY

Use metadata to track data lineage



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## ACTIVITY

# DESIGN THE STAGING AREA

Set up staging tables for intermediate data storage

## POSSIBLE PROBLEMS

# DATA LOSS DURING TRANSFORMATION

Implement robust staging tables with error handling



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## ACTIVITY

# MANAGE SURROGATE KEY PIPELINE

Ensure consistent assignment of surrogate keys

## POSSIBLE PROBLEMS

### KEY COLLISION

Use a centralized surrogate key generator



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## ACTIVITY

# ENFORCE REFERENTIAL INTEGRITY

Ensure relationships between fact and dimension tables are maintained

## POSSIBLE PROBLEMS

# ORPHANED RECORDS

Implement foreign key constraints and validation checks



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## ACTIVITY

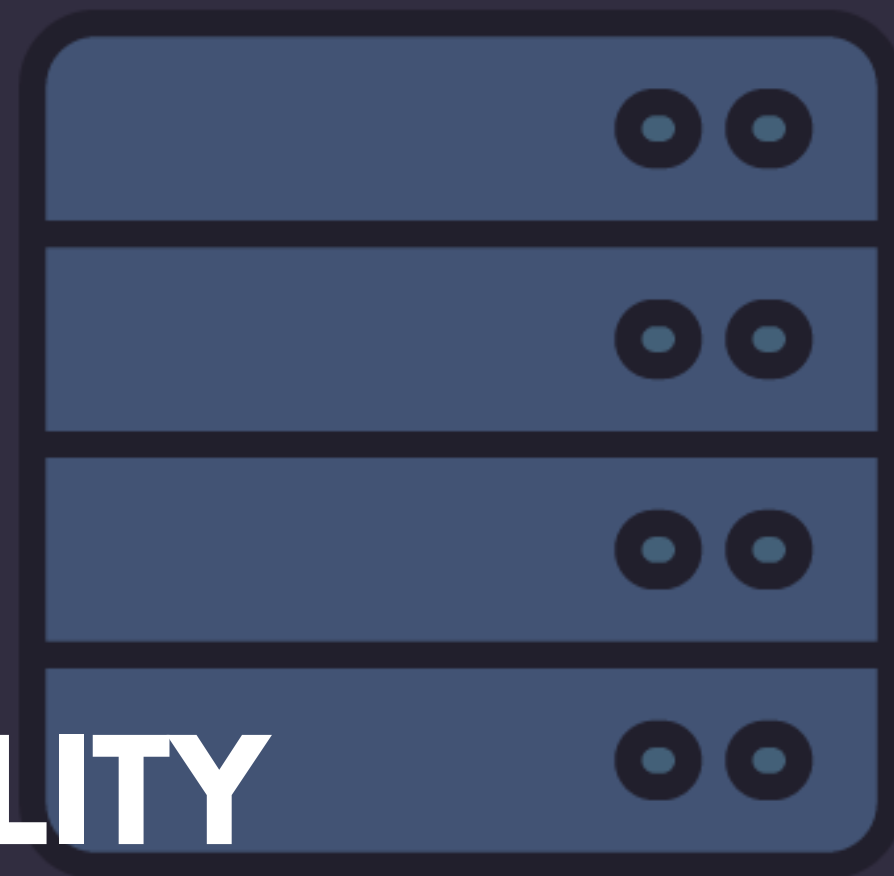
# IMPLEMENT AUDITING MECHANISMS

Track changes and data movement for compliance

## POSSIBLE PROBLEMS

# LACK OF TRACEABILITY

Use audit tables and logs



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## ACTIVITY

# HANDLE MISSING DATA

Implement strategies for dealing with missing or incomplete data

## POSSIBLE PROBLEMS

### DATA GAPS

Use default values, imputation, or flagging mechanisms



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## ACTIVITY

# PERFORM DATA PROFILING

Analyze source data for quality and completeness

## POSSIBLE PROBLEMS

# POOR DATA QUALITY

Use data profiling tools and techniques



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## ACTIVITY

# DESIGN ERROR HANDLING PROCESSES

Implement processes to handle data errors during transformation

## POSSIBLE PROBLEMS

### DATA CORRUPTION

Use error tables and retry mechanisms



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## ACTIVITY

# MANAGE DATA SECURITY

Ensure sensitive data is protected during transformation and storage

## POSSIBLE PROBLEMS

### DATA BREACHES

Implement encryption and access controls



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## ACTIVITY

### IMPLEMENT DATA VALIDATION RULES

Ensure data meets quality standards before loading into target tables

## POSSIBLE PROBLEMS

### INVALID DATA

Use validation scripts and rules



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## ACTIVITY

# IMPLEMENT BUSINESS RULES

Apply business logic during data transformation

## POSSIBLE PROBLEMS

# INCONSISTENT BUSINESS LOGIC

Centralize business rules in a metadata-driven framework



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## ACTIVITY

# HANDLE DATA LATENCY REQUIREMENTS

Ensure timely delivery of data to end users

## POSSIBLE PROBLEMS

### DELAYED DATA AVAILABILITY

Optimize ETL processes for performance



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## ACTIVITY

# MANAGE HISTORICAL DATA

Implement strategies for archiving and retrieving historical data

## POSSIBLE PROBLEMS

### DATA OVERLOAD

Use partitioning and archiving strategies



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## ACTIVITY

# DESIGN METADATA REPOSITORY

Implement a repository for  
managing metadata

## POSSIBLE PROBLEMS

# LACK OF METADATA

Use metadata tools and  
documentation



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## ACTIVITY

# MANAGE METADATA

Ensure metadata is up-to-date and accurate

## POSSIBLE PROBLEMS

# OUTDATED METADATA

Regularly update metadata repository



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## ACTIVITY

# IMPLEMENT METADATA-DRIVEN ETL

Use metadata to drive ETL processes

## POSSIBLE PROBLEMS

# HARDCODED ETL LOGIC

Use dynamic ETL based on metadata



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## ACTIVITY

# CREATE SURROGATE KEY MAPPINGS

Map natural keys to surrogate keys in staging

## POSSIBLE PROBLEMS

### KEY MISMATCHES

Implement lookup tables for key mapping



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## ACTIVITY

# HANDLE PARENT-CHILD RELATIONSHIPS

Implement strategies for hierarchies with parent-child relationships

## POSSIBLE PROBLEMS

# COMPLEX HIERARCHIES

Use recursive queries and bridge tables



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## ACTIVITY

# IMPLEMENT RECURSIVE RELATIONSHIPS

Design recursive structures for self-referencing hierarchies

## POSSIBLE PROBLEMS

# RECURSIVE QUERIES

Use recursive CTEs and hierarchy functions



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## ACTIVITY

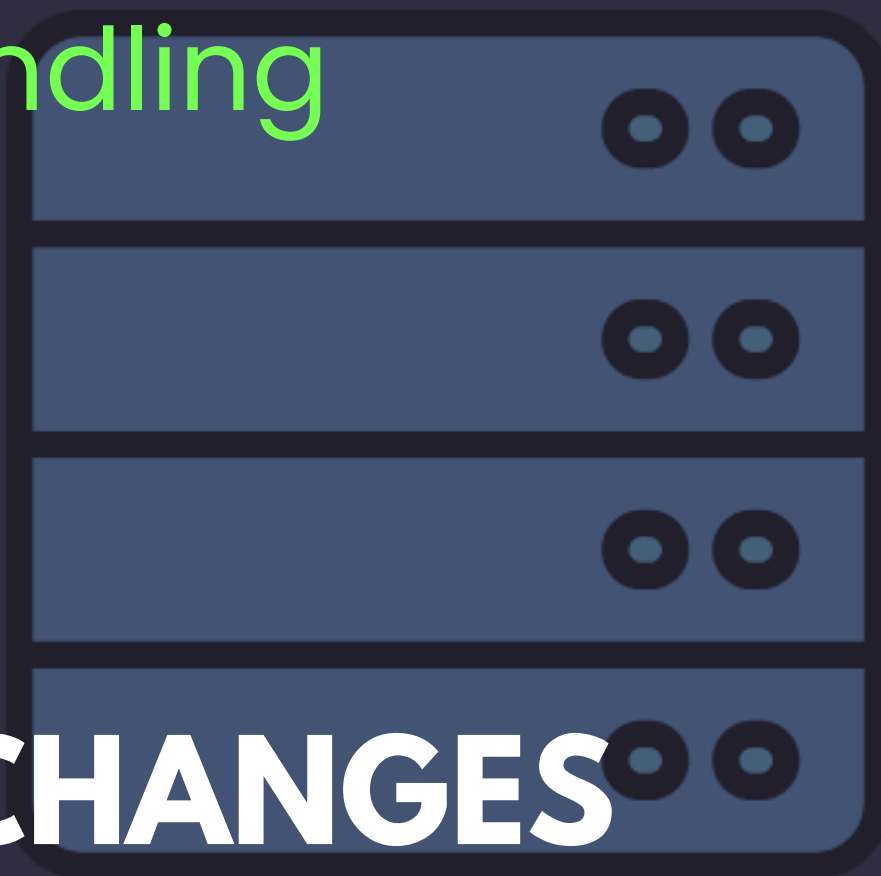
# DESIGN SLOWLY CHANGING DIMENSION (SCD) PROCESSES

Define processes for handling  
different SCD types

## POSSIBLE PROBLEMS

# HISTORICAL DATA CHANGES

Implement Type 1, 2, 3 SCDs as  
needed



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## ACTIVITY

# AUTOMATE SCD UPDATES

Automate the update of SCDs using ETL tools

## POSSIBLE PROBLEMS

# MANUAL SCD UPDATES

Use ETL tools with built-in SCD ✨  
handling



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## ACTIVITY

# TRACK SCD CHANGES

Implement audit trails for SCD changes

## POSSIBLE PROBLEMS

# LACK OF CHANGE TRACKING

Use history tables and change logs



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## ACTIVITY

# IMPLEMENT LATE-ARRIVING FACTS HANDLING

Design processes to handle late-arriving facts

## POSSIBLE PROBLEMS

### MISALIGNED DATA

Adjust fact processing to accommodate late arrivals



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## ACTIVITY

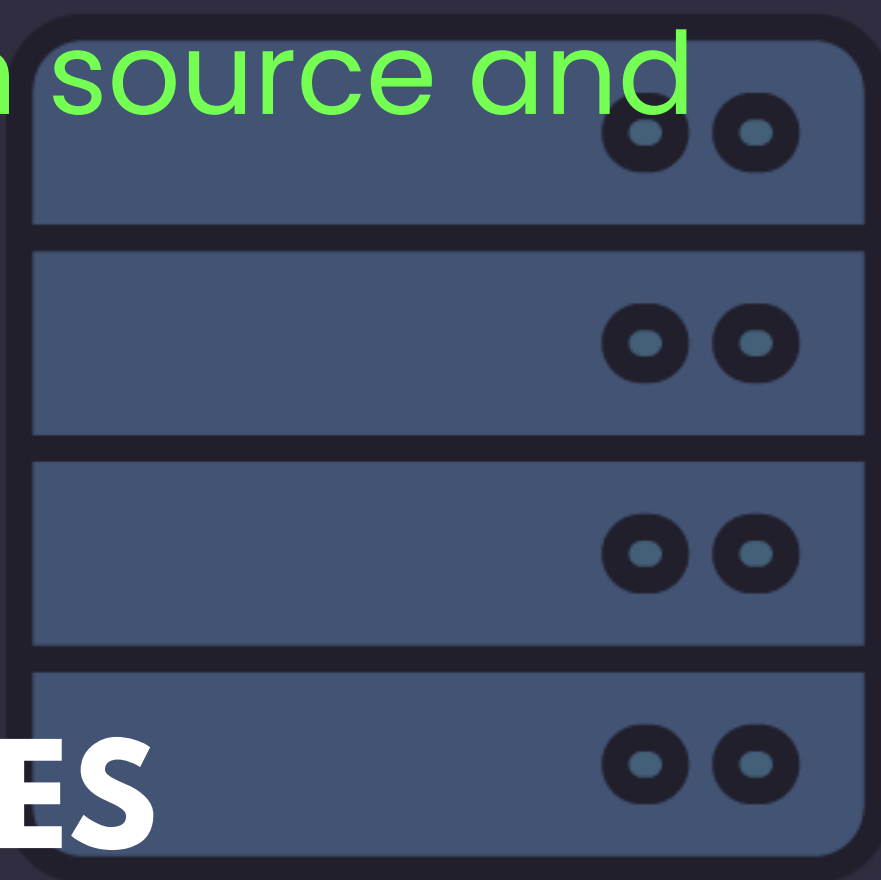
# DESIGN DATA RECONCILIATION PROCESSES

Reconcile data between source and target systems

## POSSIBLE PROBLEMS

# DATA DISCREPANCIES

Implement reconciliation reports and processes



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## ACTIVITY

# MANAGE DATA QUALITY

Implement data quality checks and validations

## POSSIBLE PROBLEMS

# POOR DATA QUALITY

Use data quality tools and rules



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## ACTIVITY

# DESIGN SURROGATE KEY MANAGEMENT SYSTEM

Centralize the management of surrogate keys

## POSSIBLE PROBLEMS

### KEY CONFLICTS

Use a surrogate key management system



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## ACTIVITY

# HANDLE NATURAL KEY CHANGES

Implement strategies for changing natural keys

## POSSIBLE PROBLEMS

# KEY CHANGES

Use surrogate keys to maintain consistency



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## ACTIVITY

# MANAGE KEY TRANSFORMATIONS

Transform keys during ETL processes

## POSSIBLE PROBLEMS

### KEY MISMATCHES

Use lookup and mapping tables



21.3

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## ACTIVITY

# DESIGN MULTI-VALUED DIMENSIONS

Implement structures for  
dimensions with multiple values

## POSSIBLE PROBLEMS

## COMPLEX QUERIES

Use bridge tables or flatten  
structures



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## ACTIVITY

# IMPLEMENT BRIDGE TABLES FOR MANY-TO-MANY RELATIONSHIPS

Design bridge tables for complex relationships

## POSSIBLE PROBLEMS

### RELATIONSHIP MANAGEMENT

Use bridge tables to simplify queries



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## ACTIVITY

# OPTIMIZE BRIDGE TABLE PERFORMANCE

Index and partition bridge tables for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing and partitioning strategies ✨



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## ACTIVITY

# IMPLEMENT HIERARCHIES WITHIN DIMENSIONS

Design hierarchical structures  
within dimensions

## POSSIBLE PROBLEMS

# COMPLEX HIERARCHIES

Use hierarchical keys and paths



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## ACTIVITY

# HANDLE UNBALANCED HIERARCHIES

Implement strategies for unbalanced hierarchies

## POSSIBLE PROBLEMS

# COMPLEX STRUCTURES

Use padding and bridge tables



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## ACTIVITY

# DESIGN SNOWFLAKE SCHEMAS

Implement snowflake schemas for normalized dimensions

## POSSIBLE PROBLEMS COMPLEX SCHEMAS

Use snowflake schemas where normalization is beneficial



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## ACTIVITY

# IMPLEMENT ROLE-PLAYING DIMENSIONS

Design dimensions that can play multiple roles

## POSSIBLE PROBLEMS

# REDUNDANT DIMENSIONS

Create views to represent role-playing dimensions



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## ACTIVITY

# OPTIMIZE ROLE-PLAYING DIMENSION PERFORMANCE

Index role-playing dimensions for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing strategies



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## ACTIVITY

# MANAGE ROLE-PLAYING DIMENSION UPDATES

Synchronize updates across role-playing dimensions

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update synchronization processes



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## ACTIVITY

# IMPLEMENT DEGENERATE DIMENSIONS

Capture transactional identifiers without corresponding dimension tables

## POSSIBLE PROBLEMS

### COMPLEX QUERIES

Use degenerate dimensions for invoice numbers, transaction IDs



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## ACTIVITY

# OPTIMIZE DEGENERATE DIMENSION PERFORMANCE

Index degenerate dimensions for  
performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing strategies



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## ACTIVITY

# MANAGE DEGENERATE DIMENSION UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# DESIGN JUNK DIMENSIONS

Consolidate low-cardinality flags and indicators into a single dimension

## POSSIBLE PROBLEMS

# EXCESSIVE DIMENSIONS

Combine attributes into a single junk dimension



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## ACTIVITY

# OPTIMIZE JUNK DIMENSION PERFORMANCE

Index junk dimensions for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing strategies



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## ACTIVITY

# MANAGE JUNK DIMENSION UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# HANDLE RAGGED HIERARCHIES

Implement hierarchies with uneven levels (e.g., geographic hierarchies)

## POSSIBLE PROBLEMS

# COMPLEX STRUCTURES

Use bridge tables to manage  
ragged hierarchies



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## ACTIVITY

# OPTIMIZE RAGGED HIERARCHY PERFORMANCE

Index ragged hierarchies for  
performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing strategies



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## ACTIVITY

# MANAGE RAGGED HIERARCHY UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# DESIGN COMMON AGGREGATES

Identify and design commonly used aggregates

## POSSIBLE PROBLEMS

# AGGREGATION INACCURACIES

Implement validation checks for aggregate data



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## ACTIVITY

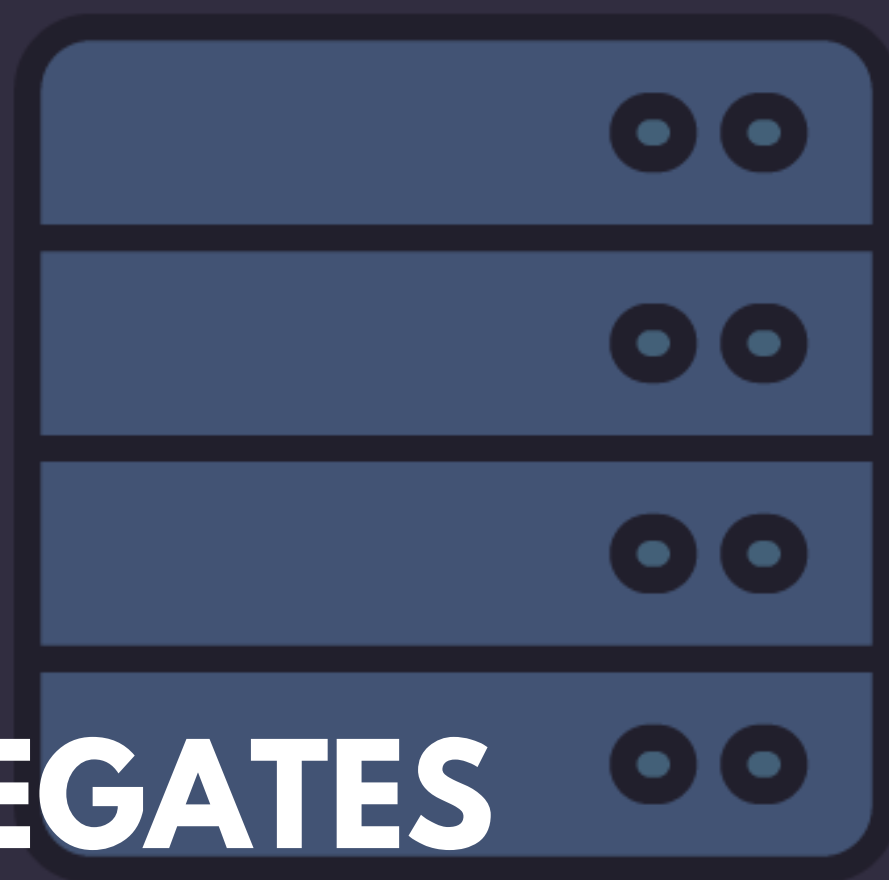
# IMPLEMENT AGGREGATE TABLES

Create summary tables to store aggregated data

## POSSIBLE PROBLEMS

# REDUNDANT AGGREGATES

Ensure aggregates add business value



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## ACTIVITY

# OPTIMIZE AGGREGATE TABLE PERFORMANCE

Index and partition aggregate tables for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing and partitioning strategies ✨



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## ACTIVITY

# DESIGN SNAPSHOT FACT TABLES

Capture data at regular intervals  
(e.g., end-of-month balances)

## POSSIBLE PROBLEMS

### DATA VOLATILITY

Use snapshot tables for periodic data



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## ACTIVITY

# OPTIMIZE SNAPSHOT FACT TABLE PERFORMANCE

Index and partition snapshot tables  
for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing and partitioning ✨  
strategies



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## ACTIVITY

# MANAGE SNAPSHOT FACT TABLE UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

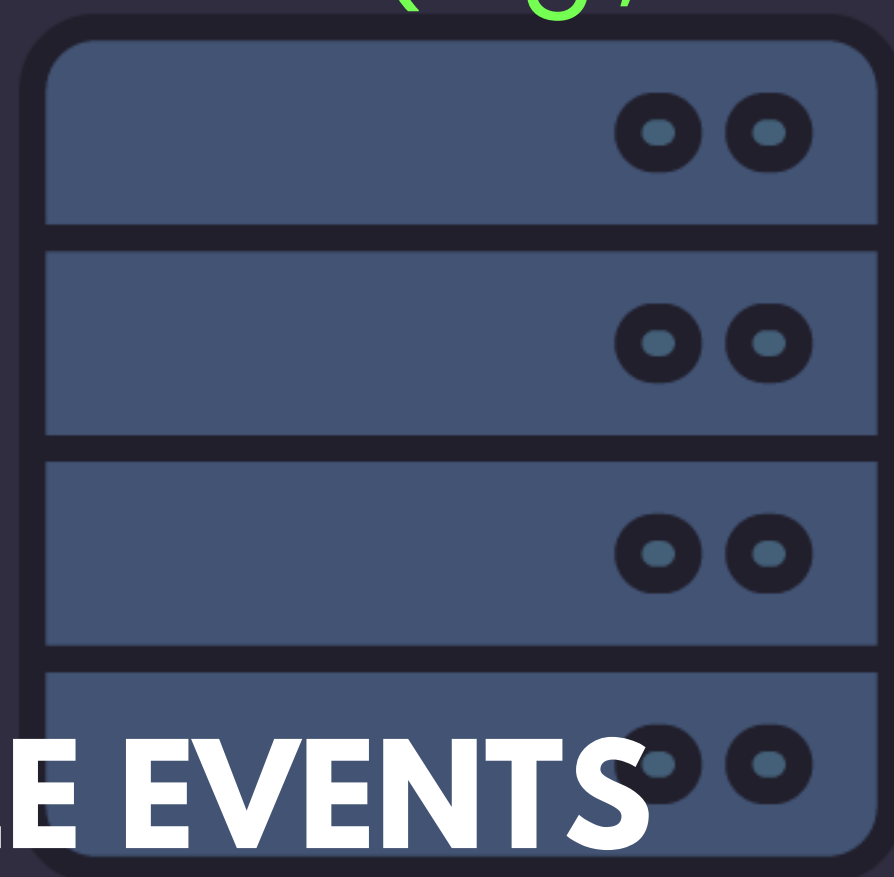
# DESIGN ACCUMULATING SNAPSHOT FACT TABLES

Track the lifecycle of an event (e.g.,  
order processing)

## POSSIBLE PROBLEMS

# COMPLEX LIFECYCLE EVENTS

Use accumulating snapshots for  
tracking progress



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## ACTIVITY

# OPTIMIZE ACCUMULATING SNAPSHOT FACT TABLE PERFORMANCE

Index and partition accumulating  
snapshot tables for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing and partitioning ✨  
strategies



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## ACTIVITY

# MANAGE ACCUMULATING SNAPSHOT FACT TABLE UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# IMPLEMENT LATE-ARRIVING FACTS HANDLING

Design processes to handle late-arriving facts

## POSSIBLE PROBLEMS

### MISALIGNED DATA

Adjust fact processing to accommodate late arrivals



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## ACTIVITY

# OPTIMIZE LATE-ARRIVING FACTS PERFORMANCE

Index and partition tables to handle late-arriving facts

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing and partitioning strategies ✨



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## ACTIVITY

# MANAGE LATE-ARRIVING FACTS UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# IMPLEMENT BRIDGE TABLES

Manage many-to-many relationships (e.g., multiple roles for an employee)

## POSSIBLE PROBLEMS

# COMPLEX RELATIONSHIPS

Use bridge tables to resolve many-to-many relationships



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## ACTIVITY

# OPTIMIZE BRIDGE TABLE PERFORMANCE

Index and partition bridge tables for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing and partitioning strategies ✨



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## ACTIVITY

# MANAGE BRIDGE TABLE UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# IMPLEMENT SLOWLY CHANGING DIMENSIONS (SCD)

Track changes in dimension data  
over time

## POSSIBLE PROBLEMS

### DATA HISTORY LOSS

Implement SCD Type 1, Type 2, and  
Type 3 as needed



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## ACTIVITY

# OPTIMIZE SCD PERFORMANCE

Index and partition SCD tables for performance

## POSSIBLE PROBLEMS

# SLOW QUERIES

Use indexing and partitioning strategies ✨



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## ACTIVITY

# MANAGE SCD UPDATES

Ensure updates are propagated correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# IMPLEMENT SURROGATE KEYS

Assign surrogate keys to maintain unique identities for dimension records

## POSSIBLE PROBLEMS

# NATURAL KEY CHANGES

Use surrogate keys to avoid issues with changing natural keys



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## ACTIVITY

# OPTIMIZE SURROGATE KEY PERFORMANCE

Index surrogate key columns for performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing strategies



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## ACTIVITY

# MANAGE SURROGATE KEY UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# IMPLEMENT HIERARCHIES WITHIN DIMENSIONS

Design hierarchical structures  
within dimensions

## POSSIBLE PROBLEMS

# COMPLEX HIERARCHIES

Use hierarchical keys and paths



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## ACTIVITY

# OPTIMIZE HIERARCHICAL DIMENSION PERFORMANCE

Index hierarchical dimensions for  
performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing strategies



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## ACTIVITY

# MANAGE HIERARCHICAL DIMENSION UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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## ACTIVITY

# IMPLEMENT DEGENERATE DIMENSIONS

Capture transactional identifiers without corresponding dimension tables

## POSSIBLE PROBLEMS COMPLEX QUERIES

Use degenerate dimensions for invoice numbers, transaction IDs



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## ACTIVITY

# OPTIMIZE DEGENERATE DIMENSION PERFORMANCE

Index degenerate dimensions for  
performance

## POSSIBLE PROBLEMS

### SLOW QUERIES

Use indexing strategies



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## ACTIVITY

# MANAGE DEGENERATE DIMENSION UPDATES

Ensure updates are propagated  
correctly

## POSSIBLE PROBLEMS

# DATA INCONSISTENCIES

Implement update processes ✨



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Thank  
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